A-A-53388
14 March 1988

COMMERCIAL ITEM DESCRIPTION

IMMPRESSION MATERIAL, DENTAL (Alginate)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers two types of dental alginate impression materials used for taking oral impressions:

Type I. - Fast Set.
Type II. - Regular (Normal) Set.

Salient characteristics:

The impression material shall be in accordance with American Dental Association (ADA) Specification No. 18 with changes to setting time, permanent deformation, compressive strength and the following additional requirements as specified below:

Setting time: From the start of mixing, initial setting time (loss of surface stickiness) shall be 1 minute 45 seconds ± 15 seconds for Type I and 2 minutes 45 seconds ± 15 seconds for Type II.

Permanent deformation. The permanent deformation shall be not more than 2.25 percent after 12 percent strain is applied for 30 seconds.

Compressive strength. Compressive strength shall be 5000 grams per square centimeter, minimum.

Odor and flavor. In addition to the ADA requirement, the odor and flavor of the material shall be evaluated by a test panel as specified in test section under odor and flavor.

Accessories. A powder proportioner, suitable for scooping the alginate powder, and a water proportioner shall be furnished in an envelope or container and shall be enclosed in a unit container. The proportioners shall be made of rigid plastic. The powder proportioner shall accommodate one scoop of powder and the water proportioner shall accommodate between 40 and 60 milliliters.

AMSC N/A FSC 6520

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The water proportioner shall be graduated to show the water levels necessary for use in either a "two-scoop system" (manufacturer's instructions describe a one-scoop impression, a two-scoop impression and a three-scoop impression). The powder and water proportioners shall be suitable for use in mixing the following types of impressions:

Two-scoop system

Powder	Water	Impression
One-scoop	Half measure	Single partial lower or upper impression
Two-scoop	Full measure	Full upper or lower impression

Three-scoop system

Powder	Water	Impression
One-scoop	1/3 measure	Single partial lower or upper impression
Two-scoop	2/3 measure	Single full lower or medium upper impression
Three-scoop	Full measure	Large full upper impression

Amount of material in unit container. The volume of material in unit container, when mixed in the proportions recommended by the manufacturer and allowed to set in a suitable mass, shall be sufficient to make a minimum of thirty impressions of 46 ml each.

Instructions for use: In addition to the requirements specified in ADA Specification No. 18, instructions shall include:

(1) water temperature to be used to obtain a 73 degree Celsius (^{o}C) mix temperature to compensate for ambient room temperatures of 18°, 23°, 24°, 27°, 30° and 33°C, (2) method of conditioning the powder prior to scooping, such as rotating or shaking the unit container and (3) method of preparing the patient's mouth prior to introducing the impression material.

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Reproducibility. The alginate material, when prepared in accordance with manufacturer's directions, shall impart a smooth, non-chalky surface and separate cleanly from a hardened gypsum cast when tested as specified in test section under reproducibility. The alginate shall be compatible with both type I, high density and type II, low density hardened dental stone of Federal Specification U-S-746.

Fineness. The material shall pass a number 20 sieve which conforms to Federal Specification RR-S-366 when tested as specified in test section under fineness.

Dust Level. The dust level shall be less than 10 micrograms per liter of air per minute and shall be determined as specified in test section under dust level.

Withdrawal Properties. From the start of mixing, withdrawal time for a finished impression shall be not greater than 3 minutes for type I and not greater than 4 minutes for type II when tested as specified in test section under withdrawal properties.

Compatibility with Gypsum. A gypsum cast poured against the alginate shall reproduce the 0.050 mm line for the full width of the ring when viewed under ten power (10X) magnification.

Consistency. The alginate shall not run out of a metal impression tray when tested as specified in test section under consistency.

Stability. The shelf life of the impression material shall be 24 months from date of manufacture and no less than 20 months shall remain from date of delivery to the Government when stored at 15°-30°C. When tested after 24 months from date of manufacture, the compressive strength shall be no less than 4000 grams per square centimeter and the setting time shall not vary by more than 30 seconds from the original requirements.

ADA Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the impression material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 18 for dental alginate impression materials. Compliance with the certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

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Tests.

Standard testing conditions. In addition to the requirements specified in American Dental Association Specification No. 18, the mix temperature shall be 23.0 ± 0.2°C for tests requiring a mixture of the alginate material.

Odor and flavor. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) units of issue of the finished material, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of odor and flavor. The remaining samples will be used by the cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the odor and flavor requirement. Approval as to odor and flavor of any sample submitted by the offeror will not constitute approval of the sample as to any other requirements of this specification. The requirement for submission of samples for use in determining compliance with the odor and flavor requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the odor and flavor requirement. An odor and taste panel consisting of 10 members shall be used to determine acceptability of samples. Samples shall be prepared for testing, coded, served to panel members under controlled conditions, e.g., all samples shall be of the same amount, and served at the same temperature; each panel member shall receive an equal number of samples; the order of serving shall be varied among panel members; an interval of at least five (5) minutes shall elapse between successive samples and panel members shall rinse their mouths with water (room temperature) after each sample; panel members shall test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the Odor and Flavor Standard, 1/ when determined by the panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

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1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ The Odor and Flavor Standard is available upon separate request to the contracting officer.

The sample is considered equal to the odor and flavor standard if its average (calculated to the nearest tenth) is not more than 0.9 units below the average (calculated to the nearest tenth) of the odor and flavor standard.

Reproducibility. A mix of the alginate material shall be made in accordance with the manufacturer's instructions and a clinical impression shall be taken of the maxillary jaw. Using the manufacturer's procedure recommended for preparing the impression for the cast, pour artificial stone (in compliance with Federal Specification U-S-746) against the impression and, when hardened, examine for cast properties specified previously in reproducibility section.

Fineness. The material shall pass a number 20 sieve when mechanically vibrated in a Ro-tap for 10 minutes.

Dust level. The dust level shall be determined by collecting the weight of airborne particles on a filter paper, under 2.5 to 2.8 mm of vacuum and 20 to 22 liters per minute of air flow, and calculating the weight in micrograms per liter. A suitable air sampler shall be used, such as a Gelman Sentry Portable Air Sampler, with a 47 mm diameter 0.45 micrometer metricil filter. The filter paper shall be weighed on a suitable 200 gram capacity analytical balance. The air shall be tested for dust level of an alginate material for a total of 5 minutes, consisting of ten 30 second runs as follows: The closed container of alginate impression powder shall be inverted 3 times prior to each 30 second run. The container shall be opened quickly, and placed so that the filter shall be as close to the center of the container as possible and 1 inch above the top surface of the alginate powder. A blank run of 1-1/2 minutes for the dust level in the atmosphere shall be made and subtracted from the dust level results for the alginate material. Two determinations of the alginate dust level shall be made on each lot or control number, and the average of the results shall be reported as the dust level of the material.

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Withdrawal. A penetrometer, with a 1.5 mm ball point needle and an arm weight of 50 gm shall be used for this test. A mix of the alginate material, made in accordance with the manufacturer's instructions, shall be placed in a Petri dish on a penetrometer plate. The initial setting time shall be determined by noting loss of stickiness to a dry finger, timed from the start of the mix and tested at 5 second intervals. As soon as the mix sets, the needle shall be adjusted to a firm contact with the top surface of the set alginate which shall be the zero reference on the indicator. Fifteen seconds after the initial set, and each fifteen seconds thereafter, the needle shall be released for five seconds, and the penetration determined by means of the indicator reading. After each test, the needle shall be returned to the zero reference and the Petri dish moved so that a new spot will be tested. This test shall be continued at fifteen second intervals until the needle drops no more than 1.5 mm. The time at which this occurs, measured from the start of mix, shall be the withdrawal time.

Consistency. An impression tray shall be filled with impression material, as prepared according to manufacturer's instructions. The impression material shall not run out of the tray when inverted for 15 seconds.

Unit. Can (CN). One can containing one pound of impression material as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product for inspection purposes shall be one can of impression material. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-01-137-8472	IMPRESSION MATERIAL, Dental, Hydrocolloid Alginate Type, Regular Set, Low Dust, 1 lb
6520-01-137-8473	IMPRESSION MATERIAL, Dental, Hydrocolloid Alginate Type, Fast Set, Low Dust, 1 lb

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-NPQAS

Project No. 6520-2049

Location: NCR Disk #4

06049

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53388	IMMPRESSION MATERIAL, DENTAL (Alginate)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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